1 Preliminaries

This chapter develops the necessary preliminaries for what follows by recalling the notions of principal bundles, classifying spaces, and characteristic classes. We assume that the reader is familiar with the concept of a fiber bundle with structure group. A classical reference for this is, for example, [?]. Moreover, we presuppose some basic familiarity with homotopy and homology theory. The first section, on principal bundles, draws mainly on [?] and [?]. In the second section, on classifying spaces, we follow [?]. Finally, the third section, on characteristic classes, is based primarily on [?], with additional material drawn from [?] and [?].

1.1 Principal Bundles and Associated Bundles

Definition 1.1. Let G be a topological group. A G-principal bundle is a fiber bundle with fiber G and structure group G acting by left translations.

The total space of a G-principal bundle $P \xrightarrow{\pi} B$ carries a natural right action of G defined as follows. Around $y \in P$, choose a local trivialisation

$$\phi\colon U \times G \to \pi^{-1}(U)$$

and let $\phi^{-1}(y) = (\pi(y), h)$. Then set

$$y \cdot g := \phi(\pi(y), hg).$$

This is well defined since, by definition, G acts on itself by *left* translations under a change of local trivialisation. A morphism of principal bundles should respect this right action, thus we define:

Definition 1.2. A morphism from a G-principal bundle $P \to B$ to an H-principal bundle $P' \to B'$ is a triple (F, f, ρ) of continuous maps $F\colon P \to P'$ and $f\colon B \to B'$ together with a homomorphism $\rho\colon G \to H$, such that

$$\begin{array}{ccc} P & \xrightarrow{F} & P' \\ \downarrow & & \downarrow \\ B & \xrightarrow{f} & B' \end{array}$$

commutes and such that F is ρ-equivariant, that is $F(x \cdot g) = F(x) \cdot \rho(g)$ for all $x \in P$ and $g \in G$. If $G = H$, we assume ρ to be the identity, and if $B = B'$, we assume f to be the identity.

The condition of equivariance in the above definition is actually quite strong, as illustrated by the following proposition.

Proposition 1.3. *Any morphism of G-principal bundles over the same base space is an isomorphism.*

Proof. First suppose we are given a morphism

$$F\colon B\times G\to B\times G$$

of trivial bundles. Then by definition F is of the form

$$F(x,g)=(x,\phi_x(g))$$

for a continuous family of maps $\phi_x\colon G\to G$. By equivariance, $\phi_x(g)=\phi_x(e)g$, and thus an inverse of F is given by

$$F^{-1}(x,g):=(x,\phi_x(e)^{-1}g).$$

Since every G-principal bundle is locally trivial, this shows that we can always find local inverses in the general case. As inverses are unique, they fit together to define a global inverse. □

To any fiber bundle with structure group G, one can canonically associate a G-principal bundle by a general construction that allows one to change the fiber of a given bundle. We now recall this construction of changing the fiber. Let $p\colon E\to B$ be a fiber bundle with fiber F and structure group G and let F' be another space upon which G acts effectively. Then we can construct a new bundle as follows. Choose an open cover $(U_\alpha)_{\alpha\in A}$ of B together with local trivialisations $\phi_\alpha\colon U_\alpha\times F\to p^{-1}(U_\alpha)$. Let $\tau_{\alpha\beta}\colon U_\alpha\cap U_\beta\to G$ be the corresponding transition functions defined by

$$(\phi_\alpha^{-1}\circ\phi_\beta)(x,f)=(x,\tau_{\alpha\beta}(x)\cdot f).$$

Define

$$E':=\left(\coprod_{\alpha\in A}U_\alpha\times F'\right)\Big/\sim$$

where the equivalence relation $\sim$ is defined by

$$(b,f',\beta)\sim(b,\tau_{\alpha\beta}(b)\cdot f',\alpha)$$

and let $p'\colon E'\to B,\ [(b,f',\alpha)]\mapsto b$. Then the maps

$$\psi_\alpha\colon U_\alpha\times F'\to(p')^{-1}(U_\alpha),\ (b,f')\mapsto[(b,f',\alpha)].$$

define local trivialisations of p' with the same transition functions $\tau_{\alpha\beta}$, giving $p'\colon E'\to B$ the structure of a fiber bundle with fiber F' and structure group G. We say that this new bundle is obtained from $E\to B$ by *changing the fiber* from F to F' (with the G-action on F' being implicit). Note that we can also obtain $E\to B$ from $E'\to B$ by changing the fiber from F' back to F. Thus no information is lost in this process. Considering the special case $F'=G$ with G acting on itself by left translations, we call the bundle obtained from $E\to B$ by changing the fiber from F to G the *associated principal bundle*. Morally speaking, if we want to understand the bundle $E\to B$, it is sufficient to understand its associated principal bundle, since this contains the same information, as we have seen above. The benefit of studying principal bundles is that they are highly symmetric objects, and thus usually easier to understand. For example, many constructions simplify for principal bundles, such as the concept of universal bundles, as we will see in the next section. Another such instance is the construction for changing the fiber from above. In the case of principal bundles this can be carried out more directly and has the benefit of also allowing for non-effective group actions. This is known as the *Borel Construction*.

Lemma 1.4. *Let $p\colon P \to B$ be a G-principal bundle and F a space upon which G acts from the left. Then*
$$E := (P \times F)/\sim, \quad (y, f) \sim (y \cdot g, g^{-1}f)$$
together with the map $\pi\colon E \to B$ given by $[(y, f)] \mapsto p(y)$ is a fiber bundle with fiber F and structure group G. Moreover, if the action of G on F is effective, this bundle is isomorphic to the bundle obtained from $P \to B$ by changing the fiber from G to F using the construction outlined above.

Proof. To see that $P \times_G F$ is locally trivial, let $\psi\colon U \times G \to P$ be a local trivialisation. Then the homeomorphism

$$\psi \times \mathrm{id}_F\colon U \times G \times F \to p^{-1}(U) \times F$$

descends to a homeomorphism

$$\tilde{\kappa}\colon (U \times G \times F)/\sim \to (p^{-1}(U) \times F)/\sim = \pi^{-1}(U),$$

where the equivalence relation on $U \times G \times F$ is given by $(x, g, f) \sim (x, gh, h^{-1}f)$. Moreover, the map
$$U \times G \times F \to U \times F,\ (x, g, f) \mapsto (x, gf)$$
descends to a homeomorphism $(U \times G \times F)/\sim \to U \times F$ with inverse given by $(x, f) \mapsto [(x, e, f)]$. Under this homeomorphism, $\tilde{\kappa}$ corresponds to a homeomorphism $\kappa\colon U \times F \to \pi^{-1}(U)$, which is explicitly given by $(x, f) \mapsto [(\psi(x, e), f)]$. Since

$$\begin{tikzcd} U \times F \arrow[r, "\kappa"] \arrow[dr, "\text{proj.}"'] & \pi^{-1}(U) \arrow[d, "\pi"] \\ & U \end{tikzcd}$$

commutes, this shows that $\pi\colon E \to B$ is locally trivial. To see that, for effective G-actions, this construction agrees with the change-of-fiber construction introduced above, we compute the transition functions of the local trivialisations just defined. Let $U_\alpha, U_\beta \subseteq B$ be two trivialising neighbourhoods for $P \to B$ with local trivialisations ψ_α, ψ_β and transition function $\tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to G$. Moreover, let $\kappa_\alpha, \kappa_\beta$ be the corresponding local trivialisations for $E \to B$ as constructed above. Then we compute

$$(\kappa_\alpha^{-1} \circ \kappa_\beta)(x, f) = \kappa_\alpha^{-1}([(\psi_\beta(x, e), f)]) = (x, \tau_{\alpha\beta}(x)f).$$

Thus the statement follows from the fact that fiber bundles, where the structure group acts effectively, are uniquely determined by the transition functions of local trivialisations. □

We denote the bundle $E \to B$ obtained from the above construction by $P \times_G F \to B$. The Borel construction is natural with respect to pullbacks, as stated in the following lemma.

Lemma 1.5. *Let $p\colon P \to B$ be a G-principal bundle, F a space upon which G acts from the left and $f\colon B' \to B$ a continuous map. Then*

$$(f^*P) \times_G F \cong f^*(P \times_G F).$$

Proof. An explicit model for the pullback bundle is

$$f^*P := \{(y, b') \in P \times B' \mid p(y) = f(b')\}.$$

Then

$$(f^*P) \times_G F = \{(y, b', f) \in P \times B' \times F \mid p(y) = f(b')\}/\sim,$$

for $(y, b', f) \sim (y \cdot g, b', g^{-1} \cdot f)$. Moreover,

$$\begin{aligned} f^*(P \times_G F) &= \{([y, f], b') \in P \times_G F \times B' \mid p(y) = f(b')\} \\ &\cong \{(y, f, b') \in P \times F \times B' \mid p(y) = f(b')\}/\sim \end{aligned}$$

for $(y, f, b') \sim (y \cdot g, g^{-1} \cdot f, b')$. Clearly these are isomorphic bundles. □

Remark 1.6. We remark that the action of G on F in the above lemma need not be effective. If we restrict ourselves to effective actions, a more general naturality result holds for the change-of-fiber construction of arbitrary bundles. Namely, first pulling a bundle back under some continuous map and then changing the fiber amounts to the same as first changing the fiber and then pulling back. Comparing the transition functions of the local trivialisations in both cases shows that they do in fact agree.

Let us look at some concrete examples for the theory developed above.

Example 1.7. (i) Let $\pi \colon E \to B$ be a real rank n vector bundle over a paracompact base space B. Choosing a bundle metric gives $E \to B$ the structure of a fiber bundle with fiber $\mathbb{R}^n$ and structure group $\mathrm{O}(n)$. The associated principal bundle is explicitly given by the *orthonormal frame bundle* $\mathrm{O}(E)$ *of* E, whose fiber over a point $x \in B$ is the space of all ordered orthonormal bases of E_x. The topology on $\mathrm{O}(E)$ is the subspace topology $\mathrm{O}(E) \subseteq E \times \ldots \times E$. Local trivialisations of the canonical projection map $p \colon \mathrm{O}(E) \to B$ can be defined as follows. Choose an orthonormal local frame $e_1, \ldots, e_n$ of E defined over an open set $U \subseteq B$. Then a trivialisation $\phi \colon U \times \mathrm{O}(n) \to p^{-1}(U)$ is given by

$$(x, A) \mapsto \left(\sum_{i=1}^{n} a_{i1} e_i(x), \ldots, \sum_{i=1}^{n} a_{in} e_i(x) \right),$$

where $A = (a_{ij})$. So $\phi(x, A)$ is the basis of $E_x = \pi^{-1}(x)$, whose coordinates in the frame (e_i) are precisely the columns of the matrix A. The transition functions of these trivialisations agree with the transition functions of the trivialisations of E. Since the transition functions uniquely determine the bundle, we conclude that $\mathrm{O}(E)$ is in fact isomorphic to the associated principal bundle as defined at the beginning of this section. Analogously, if $E \to B$ is a complex rank n vector bundle, choosing a hermitian metric on E allows us to reduce the structure group to $\mathrm{U}(n)$, and the corresponding associated principal bundle is the *unitary frame bundle* $\mathrm{U}(E)$.

(ii) Let $p \colon \tau_1^n \to \mathbb{C}P^n$ be the tautological complex line bundle equipped with the canonical hermitian metric. Since this bundle is of rank 1, its unitary frame bundle is simply the subspace $U(\tau_1^n) \subseteq \tau_1^n$ of all vectors of norm 1, together with the restriction of p to this subspace. Recall that

$$\tau_1^n = \{(L, z) \in \mathbb{C}P^n \times \mathbb{C}^{n+1} \mid z \in L\}.$$

So, explicitly,

$$U(\tau_1^n) = \{(L, z) \in \tau_1^n \mid |z| = 1\}.$$

Consider the diffeomorphism

$$U(\tau_1^n) \to S^{2n+1}, \ (L, z) \mapsto z$$

with inverse $S^{2n+1} \to U(\tau_1^n), \ z \mapsto (\pi(z), z)$, where $\pi \colon S^{2n+1} \to \mathbb{C}P^n$ is the canonical projection. Under this diffeomorphism the restriction of p to $U(\tau_1^n)$ corresponds to π. Hence π defines a $U(1) \cong S^1$-principal bundle, which agrees with the associated principal bundle of $\tau_1^n \to \mathbb{C}P^n$.

(iii) Let $E \to B$ be an S^1-principal bundle over a smooth manifold B without boundary. We can change the fiber of E from S^1 to D^2 to obtain another bundle $DE \to B$, where S^1 acts on D^2 by rotations. By definition of the change-of-fiber construction, trivialisations $U \times D^2 \to DE$ have the same transition functions as the corresponding trivialisations of $E \to B$. Moreover, they restrict to trivialisations $U \times S^1 \to \partial DE$ of the boundary. Since the action of S^1 on D^2 restricts to the action of S^1 on itself, we see that $E \cong \partial DE$. So, in particular, the total space of any S^1-principal bundle is null-bordant. We will make use of this fact in Chapter 4.

1.2 Universal Bundles and Classifying Spaces

A natural question is which fiber bundles with specified fiber and structure group can exist over a given base space. As before, this problem turns out to be easier for principal bundles. We assume all base spaces to be CW-complexes in this section.

Definition 1.8. A G-principal bundle $E \to B$ is called *universal* if E is weakly contractible as a topological space.

The significance of this definition is established by the following theorem. We say that two morphisms of G-principal bundles are G-homotopic if they are homotopic through morphisms of G-principal bundles.

Theorem 1.9. *Let $E \to B$ be a universal G-principal bundle. Then for any other G-principal bundle $P \to X$ there exists a morphism $P \to E$ of G-principal bundles, which is unique up to G-homotopy.*

Before showing the theorem, let us discuss its implications. Firstly, we can equivalently formulate the theorem by saying that $E \to B$ is a terminal object in the homotopy category of G-principal bundles over CW-complexes and thus, in particular, uniquely determined (up to G-homotopy equivalence). In view of this, one usually writes $EG := E$ and $BG := B$, and refers to $EG \to BG$ as *the* universal G-principal bundle. The base space BG is called the *classifying space* of G. Since any bundle morphism $(F, f)\colon P \to EG$ induces a morphism $P \to f^*EG$ (and vice versa, which is true for arbitrary fiber bundles), we conclude from Proposition 1.3 that

$$P \cong f^*EG.$$

We call f a *classifying map* for P. By homotopy invariance of the pullback, we get a well-defined bijection

$$\phi_X\colon [X, BG] \to \mathcal{P}_G(X),\ [f] \mapsto [f^*EG],$$

where $\mathcal{P}_G(X)$ denotes the set of isomorphism classes of G-principal bundles over X. The existence result in Theorem 1.9 shows that ϕ_X is surjective and the uniqueness result shows that it is injective. Moreover, ϕ_X is clearly natural in X. Thus we have shown:

Corollary 1.10. *The maps ϕ_X define a natural isomorphism of contravariant functors*

$$[-, BG], \mathcal{P}_G(-)\colon \mathsf{HCW} \to \mathsf{Set},$$

where HCW *denotes the homotopy category of CW-complexes.*

Theorem 1.9 will be a simple application of the following theorem.

Theorem 1.11. *Let $P \to X$ be a G-principal bundle and $A \subseteq X$ a subcomplex. Then any morphism $P|_A \to EG$ of G-principal bundles extends to a morphism $P \to EG$ of G-principal bundles.*

The proof of this theorem is based on an inductive, cell-by-cell approach. We refer the reader to [**?**, Theorem 2.7.1].

Proof of Theorem 1.9. Applying Theorem 1.11 to $A = \emptyset$ shows the existence of a morphism $P \to EG$. Now suppose we are given morphisms $F_0, F_1 \colon P \to EG$. Consider the G-principal bundle

$$\pi \times \mathrm{id} \colon P \times [0,1] \to X \times [0,1],$$

where $\pi \colon P \to X$ is the bundle projection (this is just the pullback bundle of P under the projection $X \times [0,1] \to X$). Then F_0 and F_1 define a morphism of G-principal bundles

$$(F_0, F_1) \colon P \times \{0,1\} = (P \times [0,1])|_{X \times \{0,1\}} \to EG,$$

which by Theorem 1.11 can be extended to $P \times [0,1]$, thus yielding a G-homotopy between F_0 and F_1. □

Of course, for this theory to be of any use, we need such universal bundles to exist. The following general existence theorem was first shown by Milnor [**?**].

Theorem 1.12. *For any topological group G there exists a universal G-principal bundle $EG \to BG$, where BG is a CW-complex.*

A priori, the space BG constructed in [**?**] may not be a CW-complex. However, if we choose a CW-approximation $f \colon X \to BG$, the total space of the bundle $f^*EG \to X$ is still weakly contractible. This follows by applying the five lemma to the following diagram.

$$\begin{tikzcd}
\pi_{k+1}(X) \arrow[r] \arrow[d, "\cong"] & \pi_k(G) \arrow[r] \arrow[d, "\mathrm{id}"] & \pi_k(f^*EG) \arrow[r] \arrow[d] & \pi_k(X) \arrow[r] \arrow[d, "\cong"] & \pi_{k-1}(G) \arrow[d, "\mathrm{id}"] \\
\pi_{k+1}(BG) \arrow[r] & \pi_k(G) \arrow[r] & \pi_k(EG) \arrow[r] & \pi_k(BG) \arrow[r] & \pi_{k-1}(G)
\end{tikzcd}$$

We can now use these classification results for principal bundles to obtain analogous results for general fiber bundles by appealing to the change-of-fiber construction discussed in the preceding section.

Corollary 1.13. *Let G be a topological group acting effectively on a space F. Then for any CW-complex X the map*

$$[X, BG] \to \mathcal{F}_{F,G}(X), \quad [f] \mapsto [f^*(EG \times_G F)]$$

is a natural bijection, where $\mathcal{F}_{F,G}(X)$ denotes the set of isomorphism classes of fiber bundles with fiber F and structure group G over X.

Proof. By Corollary 1.10, the map

$$[X, BG] \to \mathcal{P}_G(X), \quad [f] \mapsto [f^*EG]$$

is a natural bijection. Moreover, since G acts effectively the map

$$\mathcal{P}_G(X) \to \mathcal{F}_{F,G}(X), \quad [P] \mapsto [P \times_G F]$$

is a bijection, which is natural by Lemma 1.5. Composing both thus yields the natural bijection

$$[X, BG] \to \mathcal{F}_{F,G}(X), \quad [f] \mapsto [(f^*EG) \times_G F] = [(f^*EG \times_G F)],$$

once more applying Lemma 1.5 in the last equality. □

We conclude this section by discussing some examples.

Example 1.14.

(i) From the long exact sequence of homotopy groups

$$\pi_k(EG) \longrightarrow \pi_k(BG) \longrightarrow \pi_{k-1}(G) \longrightarrow \pi_{k-1}(EG)$$

we conclude that $\pi_k(BG) \cong \pi_{k-1}(G)$ for $k \geq 1$. Moreover, BG is connected as the image of the connected space EG under $EG \to BG$.

(ii) The universal cover $\mathbb{R} \to S^1$ is a $\mathbb{Z}$-principal bundle, $\mathbb{Z}$ acting on itself by addition. Since $\mathbb{R}$ is contractible we obtain $B\mathbb{Z} \cong S^1$ and $E\mathbb{Z} \cong \mathbb{R}$.

(iii) The universal $O(k)$-principal bundle $EO(k) \to BO(k)$ also has an explicit description. Consider the Stiefel manifold $V_k^n \subseteq \mathbb{R}^{n\times k}$, which is defined as the compact space of orthonormal k-tuples of vectors in $\mathbb{R}^n$. Furthermore, let Gr_k^n be the Grassmannian of k-planes in $\mathbb{R}^n$, i.e. the set of k-dimensional linear subspaces of $\mathbb{R}^n$ with the topology induced by the projection map

$$p\colon V_k^n \to \mathrm{Gr}_k^n,\ (v_1, \ldots, v_k) \mapsto \mathrm{span}(v_1, \ldots, v_k).$$

Identifying a linear subspace $V \in \mathrm{Gr}_k^n$ with $\mathbb{R}^k$ yields an identification of the fiber $p^{-1}(V)$ with $O(k)$. Moreover, the identifications $V \cong \mathbb{R}^k$ can be chosen consistently in a small neighbourhood around V, providing local trivialisations for p. The transition function of two such trivialisations takes values in $O(k)$, hence p defines an $O(k)$-principal bundle. The canonical inclusion $\mathbb{R}^n \to \mathbb{R}^{n+1}$ induces inclusions $V_k^n \to V_k^{n+1}$ and $\mathrm{Gr}_k^n \to \mathrm{Gr}_k^{n+1}$. Consider the colimits over these inclusions

$$V_k^\infty := \operatorname*{colim}_n (V_k^n), \quad \mathrm{Gr}_k^\infty := \operatorname*{colim}_n (\mathrm{Gr}_k^n)$$

and the induced map $V_k^\infty \to \mathrm{Gr}_k^\infty$. One can choose trivialisations of $V_k^n \to \mathrm{Gr}_k^n$ for varying n, which fit together to induce a trivialisation in the colimit, such that $V_k^\infty \to \mathrm{Gr}_k^\infty$ defines an $O(k)$-principal bundle. We claim that V_k^∞ is weakly contractible. Consider the map

$$V_k^n \to S^{n-1},\ (v_1, \ldots, v_k) \mapsto v_1.$$

Similar arguments as above show that this map is locally trivial with fiber V_{k-1}^{n-1}. Hence we can consider the long exact sequence of homotopy groups

$$\pi_{i+1}(S^{n-1}) \longrightarrow \pi_i(V_{k-1}^{n-1}) \longrightarrow \pi_i(V_n^k) \longrightarrow \pi_i(S^{n-1})$$

So $\pi_i(V_k^n) \cong \pi_i(V_{k-1}^{n-1})$ for $i \leq n-3$. Applying this inductively yields

$$\pi_i(V_k^n) \cong \pi_i(V_1^{n-(k-1)}) = \pi_i(S^{n-k+1}) = 0$$

for $i \leq n-k-1$. Now from compactness of S^i and the fact that V_k^n is closed in V_k^{n+1}, we conclude that the image of any map $S^i \to V_k^\infty$ is contained in some finite stage V_k^n. Therefore $\pi_i(V_k^\infty) = 0$ for all $i \geq 0$, such that the universal $O(k)$-principal bundle is explicitly given by $V_k^\infty \to \mathrm{Gr}_k^\infty$.

Similarly we get an explicit description of the universal $SO(k)$-principal bundle. Let $\tilde{\mathrm{Gr}}_k^n$ be the set of tuples consisting of a k-dimensional linear subspace of $\mathbb{R}^n$ together with a chosen orientation. The topology on $\tilde{\mathrm{Gr}}_k^n$ is induced by

$$p\colon V_k^n \to \tilde{\mathrm{Gr}}_k^n,\ (v_1, \ldots, v_k) \mapsto (\mathrm{span}(v_1, \ldots, v_k), \mathrm{o}_{(v_1,\ldots,v_k)}),$$

where $\mathrm{o}_{(v_1,\dots,v_k)}$ is the unique orientation on $\operatorname{span}(v_1,\dots,v_k)$, such that $(v_1,\dots,v_k)$ is positively oriented. Arguments analogous to the above show that p defines an $SO(k)$-principal bundle. Taking the colimit, we obtain the universal $SO(k)$-principal bundle $V_k^\infty \to \tilde{\mathrm{Gr}}_k^\infty$.

(iv) Any real k-dimensional vector bundle $E \to X$ can be considered as a fiber bundle with fiber $\mathbb{R}^k$ and structure group $O(k)$. So Corollary 1.13 and the preceding example show that every such E is the pullback of the universal real k-dimensional vector bundle $V_k^\infty \times_{O(k)} \mathbb{R}^k$ under some map $X \to \mathrm{Gr}_k^\infty$. We show that this universal vector bundle also has a more explicit description, namely as the tautological bundle $\gamma_k \to \mathrm{Gr}_k^\infty$. Recall that this bundle is defined as

$$\gamma_k := \{(X,v) \in \mathrm{Gr}_k^\infty \times \mathbb{R}^\infty \mid v \in X\}$$

together with the canonical projection map to Gr_k^∞. An element in V_k^∞ is a k-tuple of orthonormal vectors in some $\mathbb{R}^n$, which we will subsequently regard as the columns of an $n \times k$ matrix. Then the map

$$V_k^\infty \times \mathbb{R}^k \to \gamma_k, \quad (A,v) \mapsto (\operatorname{im}(A), Av)$$

descends to the quotient, defining a vector bundle map $V_k^\infty \times_{O(k)} \mathbb{R}^k \to \gamma_k$. This map is clearly a linear isomorphism on fibers, hence an isomorphism of vector bundles.

Similarly, oriented real k-dimensional vector bundles are classified by maps to $\tilde{\mathrm{Gr}}_k^\infty$. We claim that the universal oriented real k-dimensional vector bundle is given by the tautological bundle $\gamma_k^{SO} \to \tilde{\mathrm{Gr}}_k^\infty$ defined by

$$\gamma_k^{SO} := \{(X, \mathrm{o}_X, v) \in \tilde{\mathrm{Gr}}_k^\infty \times \mathbb{R}^k \mid v \in X\}$$

together with the canonical projection map to $\tilde{\mathrm{Gr}}_k^\infty$. To this end consider the map

$$V_k^\infty \times \mathbb{R}^k \to \gamma_k^{SO}, \quad (A,v) \mapsto (\operatorname{im}(A), \mathrm{o}_A, Av),$$

where o_A is the unique orientation on $\operatorname{im}(A)$, such that $A\colon \mathbb{R}^k \to \operatorname{im}(A)$ is orientation preserving. Again this map descends to the quotient, defining an isomorphism $V_k^\infty \times_{SO(k)} \mathbb{R}^k \to \gamma_k^{SO}$.

(v) Completely analogous to the real case, we can also define the complex Stiefel manifolds $V_k^\infty(\mathbb{C})$ and complex Grassmannians $\mathrm{Gr}_k^\infty(\mathbb{C})$ and show that the universal $U(k)$-principal bundle is given by $V_k^\infty(\mathbb{C}) \to \mathrm{Gr}_k^\infty(\mathbb{C})$. The universal complex k-dimensional vector bundle is the tautological bundle $\tau_k \to \mathrm{Gr}_k^\infty(\mathbb{C})$.

(vi) From the special cases $k=1$ of the preceding examples we obtain

$$B\mathbb{Z}_2 \cong \mathbb{R}P^\infty, \quad BS^1 \cong \mathbb{C}P^\infty.$$

1.3 Stiefel-Whitney and Chern-Classes

Again we assume all base spaces to be CW-complexes in this section. In the preceding chapter we discussed the question of how bundles can be classified. Another fundamental question is, of course, how to determine whether two given bundles are isomorphic. As usual in topology, a partial solution to this problem is provided by constructing computable invariants. For vector bundles the most important invariants are so-called *characteristic classes*. The general definition of a characteristic class is as follows. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and write $\mathrm{Vect}_{\mathbb{K}}(X)$ for the set of isomorphism classes of $\mathbb{K}$-vector bundles over a space X.

Together with the pullback operation, $\mathrm{Vect}_{\mathbb{K}}(-)$ defines a contravariant functor $\mathsf{CW} \to \mathsf{Set}$. A characteristic class c for $\mathbb{K}$-vector bundles with values in a cohomology theory H^* is a natural transformation from the functor $\mathrm{Vect}_{\mathbb{K}}(-)$ to the functor H^*, regarded as a functor with values in Set. In more concrete terms, c assigns to each $\mathbb{K}$-vector bundle $E \to X$ a cohomology class $c(E) \in H^*(X)$, depending only on the isomorphism class of E, such that for any continuous map $f\colon X' \to X$ the following naturality condition holds:

$$c(f^*E) = f^*c(E).$$

Using the theory of classifying spaces developed above, it follows that for real (resp. complex) k-dimensional vector bundles, any characteristic class c is uniquely determined by its value on the universal bundle γ_k (resp. τ_k) discussed in Example 1.14 (iv) (resp. **??**).

For real vector bundles, the most important characteristic classes are the *Stiefel-Whitney classes*, and for complex vector bundles, the *Chern classes*. Both are characterized by a few axioms.

Definition 1.15 (Stiefel-Whitney Classes)**.** There is a unique sequence of characteristic classes $w_1, w_2, \dots$ for real vector bundles with values in $H^*(-;\mathbb{Z}_2)$, such that

(i) $w_i(E) \in H^i(X;\mathbb{Z}_2)$ for any real vector bundle $E \to X$,

(ii) $w_i(E) = 0$ for $i > \mathrm{rank}(E)$,

(iii) $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$, where $w = 1 + w_1 + w_2 + \dots$,

(iv) For the universal real line bundle $\gamma_1 \to \mathbb{R}P^\infty$, $w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^\infty;\mathbb{Z}_2) \cong \mathbb{Z}_2$.

Note that by property (ii), the sum $w = 1+w_1+w_2+\dots$ is always finite. The class $w(E)$ is called the *total Stiefel-Whitney class of* E. We sometimes write $w_0(E) := 1 \in H^0(X;\mathbb{Z}_2)$ to simplify notation.

Definition 1.16 (Chern Classes)**.** There is a unique sequence of characteristic classes $c_1, c_2, \dots$ for complex vector bundles with values in $H^*(-;\mathbb{Z})$, such that

(i) $c_i(E) \in H^{2i}(X;\mathbb{Z})$ for any complex vector bundle $E \to X$,

(ii) $c_i(E) = 0$ for $i > \mathrm{rank}(E)$,

(iii) $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$, where $c = 1 + c_1 + c_2 + \dots$,

(iv) For the universal complex line bundle $\tau_1 \to \mathbb{C}P^\infty$, $c_1(\tau_1)$ is a generator of $H^2(\mathbb{C}P^\infty;\mathbb{Z}) \cong \mathbb{Z}$ specified in advance.

Again the *total Chern class* $c = 1 + c_1 + c_2 + \dots$ is well-defined and we write $c_0(E) := 1 \in H^0(X;\mathbb{Z})$. Note that in both definitions, property (iv) uniquely determines the values of the respective characteristic classes on line bundles and, together with property (iii), prescribes their values on bundles that split as sums of line bundles. The values on arbitrary vector bundles can then be deduced from a general splitting principle [**?**, Proposition 3.3]. We refer the reader to [**?**] for a detailed proof of existence and uniqueness.

Not only are the definitions of Stiefel-Whitney and Chern classes formally quite similar, but there is also a direct connection between the two.

Proposition 1.17 ([**?**, Proposition 3.8])**.** *Regarding a complex rank n vector bundle $E \to X$ as a real rank $2n$ vector bundle, $w_{2i+1}(E) = 0$ and $w_{2i}(E)$ is the image of $c_i(E)$ under the coefficient homomorphism $H^{2i}(X;\mathbb{Z}) \to H^{2i}(X;\mathbb{Z}_2)$.*

Let us look at a few examples.

Example 1.18.

(i) Let $\iota\colon \mathbb{R}P^n \to \mathbb{R}P^\infty$ be the canonical inclusion. Then the pullback $\iota^*\gamma_1$ is isomorphic to the tautological line bundle $\gamma_1^n \to \mathbb{R}P^n$. Since ι induces an isomorphism on cohomology, we conclude from naturality that $w_1(\gamma_1^n) = \iota^* w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^n;\mathbb{Z}_2) \cong \mathbb{Z}_2$. Similarly, for the tautological complex line bundle $\tau_1^n \to \mathbb{C}P^n$, the class $c_1(\tau_1^n)$ is a generator of $H^2(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}$.

(ii) Let $E \to X$ be a trivial real vector bundle. Then $E \cong f^*E$ for a constant map $f\colon X \to X$, and thus $w_i(E) = f^*w_i(E) = 0$ for all $i \geq 1$. Similarly, all Chern classes for complex trivial vector bundles vanish.

(iii) Let $TS^n \to S^n$ be the tangent bundle of the sphere and let $\nu_n \to S^n$ be the normal bundle with respect to the standard embedding $S^n \to \mathbb{R}^{n+1}$. Since ν_n and $TS^n \oplus \nu_n$ are trivial, we conclude from the previous example and property (iii) in Definition 1.15 that

$$1 = w(TS^n \oplus \nu_n) = w(TS^n) \smile w(\nu_n) = w(TS^n) \smile 1 = w(TS^n).$$

Thus $w_i(TS^n) = 0$ for all $i \geq 1$.

The last example shows that although all Stiefel-Whitney classes of a vector bundle vanish, it need not be trivial. However, for real (resp. complex) vector bundles of rank 1, it turns out that the first Stiefel-Whitney (resp. Chern) class already determines the bundle:

Proposition 1.19 ([**?**, Proposition 3.10])**.** *The maps*

$$w_1\colon \mathrm{Vect}_{\mathbb{R}}(X) \to H^1(X;\mathbb{Z}_2)$$

and

$$c_1\colon \mathrm{Vect}_{\mathbb{C}}(X) \to H^2(X;\mathbb{Z})$$

are bijections.

While the first Stiefel-Whitney class of higher dimensional bundles does not identify their isomorphism type, it does characterize another important invariant, namely orientability:

Proposition 1.20 ([**?**, Proposition 3.11])**.** *A real vector bundle $E \to X$ is orientable if and only if $w_1(E) = 0$.*

Similarly, vanishing of the second Stiefel-Whitney class corresponds to the existence of another structure, namely that of a *spin structure*.

Definition 1.21. (i) For $n \geq 2$ the *spin group* $\mathrm{Spin}(n)$ is defined as the total space of the non-trivial two-sheeted cover $\rho\colon \mathrm{Spin}(n) \to \mathrm{SO}(n)$. If $n \geq 3$, this is the universal cover, and in the case $n = 2$ we have $\mathrm{Spin}(2) \cong \mathrm{SO}(2)$. For $n = 1$ we define $\mathrm{Spin}(1) := \mathbb{Z}_2$, the total space of the two-sheeted cover $\rho\colon \mathbb{Z}_2 \to \mathrm{SO}(1)$.

(ii) Let $E \to X$ be an oriented real vector bundle equipped with a fiber metric and let $\mathrm{SO}(E)$ be its oriented orthonormal frame bundle. A *spin structure* on E is a $\mathrm{Spin}(n)$-principal bundle $\mathrm{Spin}(E) \to X$ together with a map

$$\phi\colon \mathrm{Spin}(E) \to \mathrm{SO}(E)$$

of principal bundles, which is ρ-equivariant (compare Definition 1.2).

(iii) An orientable manifold M is called *spin* if its tangent bundle TM admits a spin structure and is called *totally non-spin* if the universal cover of M is not spin.

Remark 1.22. When $n = 1$, $\mathrm{SO}(E) \cong X$ and a spin structure is simply a two-sheeted cover of X.

Note that the existence of a spin structure does not depend on the chosen orientation or fiber metric on E, since different choices yield isomorphic oriented orthonormal frame bundles. In fact, the only obstruction to the existence of a spin structure is the second Stiefel-Whitney class, as mentioned above:

Theorem 1.23 ([**?**, Chapter 2, Theorem 1.7])**.** *An orientable real vector bundle $E \to X$ admits a spin structure if and only if $w_2(E) = 0$.*

Example 1.24. (i) It follows from the above theorem and Example 1.18 (iii) that all spheres S^n, $n \geq 1$, are spin.

(ii) The total Chern class of the tangent bundle $T\mathbb{C}P^n$ of $\mathbb{C}P^n$ can be computed to be

$$c(T\mathbb{C}P^n) = (1 + \alpha)^{n+1}$$

for a suitably chosen generator $\alpha \in H^2(\mathbb{C}P^n; \mathbb{Z})$; see [**?**, Theorem 14.10]. In particular, $c_1(T\mathbb{C}P^n) = (n+1)\alpha$. From Proposition 1.17 we see that $w_2(T\mathbb{C}P^n) = 0$ if and only if n is odd. Therefore, $\mathbb{C}P^n$ is spin if and only if n is odd.

Remark 1.25. Let $E \to X$ be a real vector bundle. If E is orientable, we have seen that this bundle is classified by a map $X \to B\mathrm{SO}(n)$. Similarly, if a spin structure exists, E is classified by a map $X \to B\mathrm{Spin}(n)$, which can be seen as follows. Define a vector bundle

$$\gamma_n^{\mathrm{Spin}} := E\mathrm{Spin}(n) \times_{\mathrm{Spin}(n)} \mathbb{R}^n$$

using Lemma 1.4, where $\mathrm{Spin}(n)$ acts on $\mathbb{R}^n$ by virtue of the covering map $\rho\colon \mathrm{Spin}(n) \to \mathrm{SO}(n)$ and the canonical action of $\mathrm{SO}(n)$. Let $\phi\colon \mathrm{Spin}(E) \to \mathrm{SO}(E)$ be a spin structure on E and let $f\colon X \to B\mathrm{Spin}(n)$ be a classifying map for $\mathrm{Spin}(E)$. From Lemma 1.5 we obtain

$$f^*\gamma_n^{\mathrm{Spin}} \cong \mathrm{Spin}(E) \times_{\mathrm{Spin}(n)} \mathbb{R}^n.$$

Moreover, the spin structure ϕ induces an isomorphism $\mathrm{Spin}(E) \times_{\mathrm{Spin}(n)} \mathbb{R}^n \to \mathrm{SO}(E) \times_{\mathrm{SO}(n)} \mathbb{R}^n \cong E$ of vector bundles by

$$[(x, v)] \mapsto [(\phi(x), v)].$$

Since this map restricts to a linear bijection on each fiber, it is an isomorphism of vector bundles.

Recall that a manifold M is orientable if and only if its tangent bundle TM restricted to any embedded loop $S^1 \subseteq M$ is trivial. Or, in other words, "TM does not contain any Moebius bands". A similar characterization can be given for spin manifolds, which is why the property of being spin is sometimes referred to as a higher dimensional analogue of orientability.

Proposition 1.26. *Let M be a connected oriented manifold of dimension at least* 5*.*

(i) M is spin if and only if for every embedded surface $S \subseteq M$ the restriction $TM|_S$ of the tangent bundle to S is trivial.

(ii) M is totally non-spin if and only if there exists an embedding $\iota\colon S^2 \to M$, such that $w_2(\iota^*TM) \neq 0$. In particular, the restriction ι^*TM of the tangent bundle to this embedded sphere is non-trivial.

Proof. Since we will not be making use of the first characterization, we shall only prove part (ii) of the proposition and refer the reader to [**?**, Proposition 2.6] for a proof of part (i). The proof closely follows [**?**]. Let $\tilde{M}$ be the universal cover of M. Suppose that $T\tilde{M}$ does not admit a spin structure. Then we conclude from Theorem 1.23 that $w_2(T\tilde{M}) \neq 0$. So, by the universal coefficient theorem, there exists a homology class $a \in H_2(\tilde{M};\mathbb{Z}_2)$ with

$$\langle w_2(T\tilde{M}), a\rangle \neq 0.$$

Once more applying the universal coefficient theorem as well as the Hurewicz theorem, we obtain an isomorphism

$$\pi_2(\tilde{M}) \otimes \mathbb{Z}_2 \to H_2(\tilde{M};\mathbb{Z}) \otimes \mathbb{Z}_2 \cong H_2(\tilde{M};\mathbb{Z}_2).$$

Explicitly, for some class $\alpha \in \pi_2(\tilde{M})$ represented by a map $f\colon S^2 \to \tilde{M}$, the above isomorphism maps $\alpha \otimes 1$ to $f_*[S^2]_{\mathbb{Z}_2}$, where $[S^2]_{\mathbb{Z}_2}$ is the $\mathbb{Z}_2$-fundamental class of S^2. Therefore we can write $a = f_*[S^2]_{\mathbb{Z}_2}$ for some $f\colon S^2 \to \tilde{M}$. Let $p\colon \tilde{M} \to M$ be the projection and define $\iota := p \circ f\colon S^2 \to M$. Since $\dim(M) \geq 5$, we may assume ι to be an embedding. We compute

$$\langle w_2(\iota^*TM), [S^2]_{\mathbb{Z}_2}\rangle = \langle f^*w_2(p^*TM), [S^2]_{\mathbb{Z}_2}\rangle = \langle w_2(T\tilde{M}), f_*[S^2]_{\mathbb{Z}_2}\rangle = \langle w_2(T\tilde{M}), a\rangle \neq 0,$$

where we used $p^*TM \cong T\tilde{M}$ and naturality of the Kronecker pairing in the second equation. Therefore $w_2(\iota^*TM) \neq 0$. Conversely, if there exists some embedding $\iota\colon S^2 \to M$ with $w_2(\iota^*TM) \neq 0$, we can lift ι along p to an embedding $\tilde{\iota}\colon S^2 \to \tilde{M}$. Again using $p^*TM \cong T\tilde{M}$ we obtain

$$w_2(\tilde{\iota}^*T\tilde{M}) = w_2(\iota^*TM) \neq 0.$$

Therefore $w_2(T\tilde{M}) \neq 0$ by naturality of w_2. □

Finally, we record another scenario in which the vanishing of all Stiefel-Whitney classes of a vector bundle is sufficient for triviality.

Lemma 1.27. *Let $E \to S$ be a real vector bundle of rank at least 3 over a two-dimensional CW-complex S. Then E is trivial if and only if $w_1(E) = w_2(E) = 0$.*

Proof. Combining Theorem 1.23 and Remark 1.25, we conclude that there exists a classifying map $f\colon S \to B\mathrm{Spin}(n)$ for E, where n is the rank of E. From Example 1.14 (i) it follows that $\pi_k(B\mathrm{Spin}(n)) = 0$ for $k \leq 2$. Since S is two-dimensional, the map f is homotopic to a constant by [**?**, Lemma 4.6], therefore $E \cong f^*\xi_{\mathrm{Spin}}$ is trivial. □

2 Main Construction

Recall the following question discussed in the Introduction.

Question. Does there exist a circle bundle $E \to M$ over an enlargeable manifold M, such that the total space E is non-enlargeable? Moreover, does there exist a 2-dimensional real vector bundle $V \to M$, such that V admits a complete metric of uniformly positive scalar curvature?

We begin this chapter with an alternative approach to the one presented in [**?**] for constructing examples addressing the first part of the question. We subsequently build on this construction to obtain an affirmative answer to the second part of the question.

2.1 Circle Bundles with Positive Scalar Curvature

In this section we construct examples of circle bundles $E \to M$ over enlargeable manifolds M in all dimensions $\dim(M) \geq 4$, such that E admits a PSC-metric. In particular E is not enlargeable by Theorem **??**. Let us first describe the general strategy behind the construction. The existence of a PSC-metric on E will be established as an application of Theorem **??**, restated here for convenience:

Theorem ??. *Let W^{n+1} be a connected oriented cobordism from M_0 to M_1, $n \geq 5$. Suppose M_1 is connected and totally non-spin, the inclusion $M_1 \to W$ induces an isomorphism on π_1, and M_0 admits a* PSC*-metric. Then M_1 also admits a* PSC*-metric. In fact, any* PSC*-metric on M_0 can be extended to a* PSC*-metric on W, which is of product form near the boundary.*

In order to apply this in our case, we make the following two observations, valid for any circle bundle $E \to M$ over a closed connected orientable manifold M.

(i) There exists an oriented nullbordism of E, namely the total space of the associated disk bundle $DE \to M$,

(ii) The inclusion $E \cong \partial DE \to DE$ induces an isomorphism on π_1 if and only if the fibers of E are contractible within E, that is the inclusion $S^1 \to E$ is trivial on π_1.

We have already observed in Example (iii) that DE is a nullbordism of E. Moreover, the structure group of $DE \to M$ acts on fibers by orientation-preserving diffeomorphisms. Thus any orientation on M gives rise to an orientation on DE, which is uniquely determined by requiring local trivialisations to be orientation preserving. For (ii), note that $DE \to M$ is a homotopy equivalence. Hence the diagram

$$\begin{tikzcd} E \arrow[r] \arrow[dr] & DE \arrow[d] \\ & M \end{tikzcd}$$

shows that $E \to DE$ induces an isomorphism on π_1 if and only if $E \to M$ does. Then the claim follows from the long exact sequence of homotopy groups:

$$\pi_1(S^1) \longrightarrow \pi_1(E) \longrightarrow \pi_1(M) \longrightarrow \pi_0(S^1).$$

With these observations in mind, we need to construct circle bundles $E \to M$ with the following properties.

(I) M is enlargeable, orientable and connected,

(II) E is totally non-spin,

(III) The inclusion $S^1 \to E$ is trivial on π_1.

We define M as the connected sum of three closed manifolds. Let N^{2n} be any closed $2n$-dimensional totally non-spin manifold, which is moreover connected and orientable. For example, take $N := \mathbb{C}P^2 \times T^{2n-4}$, which is totally non-spin by Lemma **??** and Example 1.24 (ii). Then consider

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Intuitively, the bundle $E \to M$ will now look as follows. It is trivial over the summands T^{2n} and N, and is given by the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$ over the $\mathbb{C}P^n$ summand (or rather the restriction of this bundle to $\mathbb{C}P^n - D^{2n} \subseteq M$). Schematically:

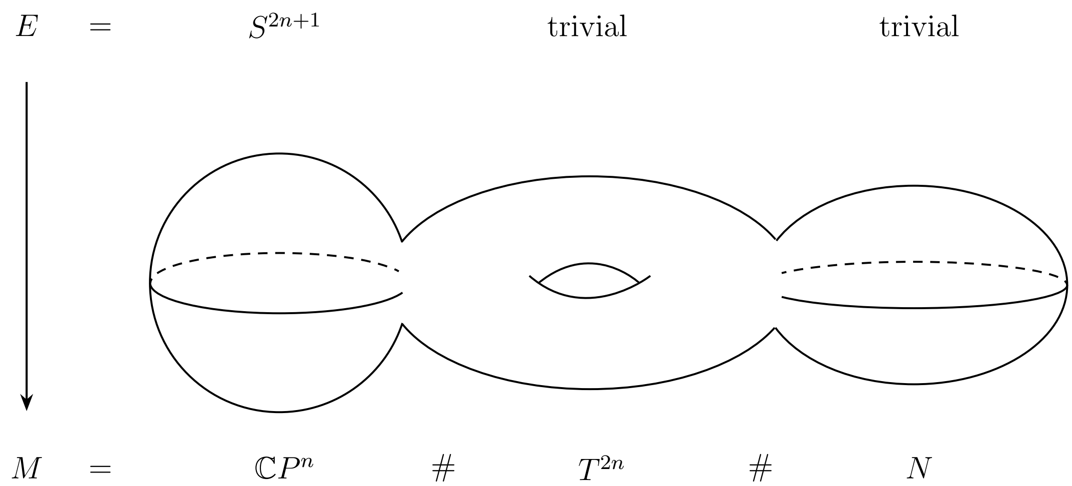

Figure 2.1: Sketch of the circle bundles constructed in this section

The presence of the torus summand then corresponds to property (I), the trivial bundle over N to property (II), and the S^{2n+1} over $\mathbb{C}P^n$ to property (III). We will now construct this bundle rigorously and subsequently discuss the proof of properties (I) - (III) in more detail. Implicitly, all homology and cohomology groups in this chapter are taken with $\mathbb{Z}$-coefficients. Recall that for manifolds M_1 and M_2 of dimension m, the cohomology of a connected sum $M_1 \# M_2$ is

$$H^k(M_1 \# M_2) \cong H^k(M_1) \oplus H^k(M_2)$$

for $k \leq m-2$. Explicitly, this isomorphism is given by the composition

$$H^k(M_1) \oplus H^k(M_2) \xrightarrow{\cong} H^k(M_1 - D^m) \oplus H^k(M_2 - D^m) \xleftarrow{\cong} H^k(M_1 \# M_2),$$

where all maps are induced by inclusions. Applying this to M, we obtain an isomorphism

$$\phi\colon H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) \to H^2(M).$$

Let $\alpha = c_1(\tau_1^n) \in H^2(\mathbb{C}P^n)$ be the first Chern class of the tautological complex line bundle $\tau_1^n \to \mathbb{C}P^n$ and set $c_1 := \phi(\alpha, 0, 0) \in H^2(M)$. By Theorem 1.19, there exists a unique complex line bundle $L \to M$ with first Chern class $c_1(L) = c_1$. Now define $E \to M$ as the S^1-principal bundle associated to this line bundle. We claim that E is of the form as described above. To this end, we will show that $L \to M$ is trivial over the summands T^{2n} and N, and is given by the restriction of $\tau_1^n \to \mathbb{C}P^n$ to $\mathbb{C}P^n - D^{2n} \subseteq M$ over the $\mathbb{C}P^n$ summand. By Example 1.7 (ii), the associated principal bundle of $\tau_1^n \to \mathbb{C}P^n$ is precisely $S^{2n+1} \to \mathbb{C}P^n$, so that the claim then follows from the naturality of the change-of-fiber construction; see Remark 1.6. To establish the statements made about $L \to M$, let

$$i\colon N - D^{2n} \to M, \quad i_1\colon N - D^{2n} \to N$$

be the inclusions. Then

$$\begin{tikzcd} H^2(M) \arrow[r, "i^*"] & H^2(N - D^{2n}) \\ H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) \arrow[u, "\cong"', "\phi"] \arrow[r, "\text{proj.}"] & H^2(N) \arrow[u, "\cong"', "i_1^*"] \end{tikzcd}$$

commutes, and therefore $c_1(i^*L) = i^*(c_1) = 0$. Again, using Theorem 1.19, we conclude that i^*L is trivial. An analogous argument shows triviality of L over the T^{2n} summand. Similarly, if we denote by

$$j\colon \mathbb{C}P^n - D^{2n} \to M, \quad j_1\colon \mathbb{C}P^n - D^{2n} \to \mathbb{C}P^n$$

the inclusions, the diagram

$$\begin{tikzcd} H^2(M) \arrow[r, "j^*"] & H^2(\mathbb{C}P^n - D^{2n}) \\ H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) \arrow[u, "\cong"', "\phi"] \arrow[r, "\text{proj.}"] & H^2(\mathbb{C}P^n) \arrow[u, "\cong"', "j_1^*"] \end{tikzcd}$$

commutes, and thus $c_1(j^*L) = j^*(c_1) = j_1^*(\alpha) = c_1(j_1^*\tau_1^n)$. Hence, applying Theorem 1.19 once more shows that j^*L and $j_1^*\tau_1^n$ are isomorphic bundles. This concludes the construction of E. Let us now verify properties (I) - (III).

Proof of property (I). M is connected and orientable, as each summand is. Moreover, since the collapse map $M \to T^{2n}$ has degree 1 and T^{2n} is enlargeable, it follows from Lemma **??** that M is also enlargeable. □

Proof of property (II). Let $i\colon N - D^{2n} \to M$ be the inclusion. We have established above that

$$i^*E \cong (N - D^{2n}) \times S^1.$$

From Lemma **??** **??** and **??**, we conclude that $(N - D^{2n}) \times S^1$ is totally non-spin. Therefore, E contains a totally non-spin submanifold of codimension 0, and so Lemma **??** **??** implies that E is also totally non-spin. □

Proof of property (III). Intuitively, we can contract the fibers over the S^{2n+1} contained in E. We can formalize this as follows. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations, and the vertical maps are induced by inclusions:

$$\begin{tikzcd} \pi_2(M) \arrow[r, "\partial_1"] & \pi_1(S^1) \arrow[r] & \pi_1(E) \\ \pi_2(\mathbb{C}P^n - D^{2n}) \arrow[u] \arrow[r, "\partial_2"] \arrow[d, "\cong"'] & \pi_1(S^1) \arrow[r] \arrow[u, "\text{id}"'] \arrow[d, "\text{id}"'] & \pi_1(j^*E) \arrow[u] \arrow[d] \\ \pi_2(\mathbb{C}P^n) \arrow[r, "\partial_3"] & \pi_1(S^1) \arrow[r] & \pi_1(S^{2n+1}) \end{tikzcd}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected, and so, by the Hurewicz Theorem,

$$\begin{tikzcd} \pi_2(\mathbb{C}P^n - D^{2n}) \arrow[r] \arrow[d, "\cong"] & \pi_2(\mathbb{C}P^n) \arrow[d, "\cong"] \\ H_2(\mathbb{C}P^n - D^{2n}) \arrow[r] & H_2(\mathbb{C}P^n) \end{tikzcd}$$

commutes, and the vertical maps are isomorphisms. Since the bottom horizontal map is an isomorphism, the same holds for the top horizontal map. We now claim that ∂_1 is surjective. For this, it suffices to show surjectivity of ∂_2, which in turn is equivalent to surjectivity of ∂_3. But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \to \pi_1(E)$$

is the zero map. □

The above construction now gives examples of circle bundles with positive scalar curvature over enlargeable manifolds M in all even dimensions $\dim(M) \geq 4$. We argue as in [**?**] to also obtain examples in odd dimensions as follows. Let $E \to M$ be as above and let $\pi\colon M \times S^1 \to M$ be the projection. Then, by Example **??**, the pullback bundle $\pi^*E \cong E \times S^1$ admits a metric of positive scalar curvature. Furthermore, the base space $M \times S^1$ is still enlargeable by Lemma **??**. Naturally, the question arises as to what happens in dimensions $\dim(M) \leq 3$. In this case, it turns out that no circle bundle $E \to M$ over an enlargeable manifold M can admit a metric of positive scalar curvature [**?**, Theorem B, Remark 2].

2.2 Vector Bundles with Uniformly Positive Scalar Curvature

In this section, we will use the preceding results to construct examples of 2-dimensional real vector bundles $V \to M$ over enlargeable manifolds M, such that V admits a complete metric of uniformly positive scalar curvature, thereby answering the second part of the question formulated at the beginning of this chapter.

Let $E \to M$ be a circle bundle as constructed in the previous section. The existence of a PSC-metric on E followed from an application of Theorem **??**. However, we did not make use of the full strength of this theorem. Not only do we obtain a PSC-metric on E, but also a PSC-metric on the nullbordism DE, which is of product form near the boundary. This observation allows for the following construction. Let g be such a PSC-metric on DE and denote the induced metric on E by g_E. Choose some collar neighbourhood $\iota\colon E \times [0, 1) \to DE$ with respect to which g is in product form, i.e.

$$\iota^* g = g_E + dt^2.$$

Moreover, consider the Riemannian manifold $(E \times [1, \infty), g_E + dt^2)$ equipped with the identity as a collar neighbourhood. We can glue this manifold with (DE, g) along the common boundary E to define another Riemannian manifold

$$(V', h) := (E \times [1, \infty) \cup_E DE, (g_E + dt^2) \cup_E g).$$

We explicitly mention that the smooth structure on V' is defined with respect to the chosen collar neighbourhoods ι and $\mathrm{id}_{E\times[1,\infty)}$, so that h is indeed smooth. Note that the metrics on both $E \times [1, \infty)$ and DE are complete. Therefore h is also complete. Furthermore, g_E is a PSC-metric, hence the scalar curvature of $g_E + dt^2$ is uniformly positive by compactness of E and Example **?? ??**. The same holds for the scalar curvature of g. Consequently, scal_h is uniformly positive as well. Now consider the map $p_0\colon E \times [1,\infty) \to M$ defined as the composition of the projection $E \times [1,\infty) \to E$ and the bundle projection $p\colon E \to M$. Together with the bundle projection $q\colon DE \to M$, we set

$$\pi := p_0 \cup_E q\colon V' \to M,$$

which we will show to be a rank 2 vector bundle. However, in general we can only expect π to be continuous, since the collar neighbourhood ι contains no information about the bundle structure on DE. To resolve this subtlety, we proceed as follows. Choose trivialisations

$$\phi_\alpha\colon U_\alpha \times S^1 \to p^{-1}(U_\alpha), \;\; \phi_\beta\colon U_\beta \times S^1 \to p^{-1}(U_\beta)$$

of $E \to M$ with smooth transition function $\tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to S^1$. By definition of the change-of-fiber construction, there are then trivialisations

$$\psi_\alpha\colon U_\alpha \times D^2 \to q^{-1}(U_\alpha), \;\; \psi_\beta\colon U_\beta \times D^2 \to q^{-1}(U_\beta)$$

of $DE \to M$ with the same transition function $\tau_{\alpha\beta}$. Let $B \subseteq \mathbb{R}^2$ be the open ball of radius 1 and consider the diffeomorphism

$$\mu\colon \mathbb{R}^2 - B \to S^1 \times [1,\infty), \; w \mapsto \left(\frac{w}{|w|}, |w|\right),$$

which is equivariant under the obvious S^1-actions on $\mathbb{R}^2 - B$ and $S^1 \times [1,\infty)$. Then we can define a trivialisation of $p_0\colon E \times [1,\infty) \to M$ by

$$\theta_\alpha\colon U_\alpha \times (\mathbb{R}^2 - B) \xrightarrow{\mathrm{id}\times\mu} U_\alpha \times S^1 \times [1,\infty) \xrightarrow{\phi_\alpha\times\mathrm{id}} p^{-1}(U_\alpha) \times [1,\infty) = p_0^{-1}(U_\alpha)$$

Analogously define $\theta_\beta\colon U_\beta \times (\mathbb{R}^2 - B) \to p_0^{-1}(U_\beta)$. By construction, the transition function of these trivialisations is again $\tau_{\alpha\beta}$. The maps θ_α and ψ_α now glue together to define a trivialisation of $\pi\colon V' \to M$ given by the *homeomorphism*

$$\xi_\alpha\colon U_\alpha \times \mathbb{R}^2 = (U_\alpha \times (\mathbb{R}^2 - B)) \cup (U_\alpha \times D^2) \xrightarrow{\theta_\alpha \cup \psi_\alpha} p_0^{-1}(U_\alpha) \cup q^{-1}(U_\alpha) = \pi^{-1}(U_\alpha).$$

Likewise, we obtain a homeomorphism $\xi_\beta\colon U_\beta \times \mathbb{R}^2 \to \pi^{-1}(U_\beta)$. From the discussion about the transition functions, we conclude

$$(\xi_\beta^{-1} \circ \xi_\alpha)(x, w) = (x, \tau_{\alpha\beta}(x) \cdot w).$$

Observe that this map is smooth. Therefore, the collection of maps constructed in the same manner as ξ_α and ξ_β defines a smooth structure on V'. This structure may not agree with the one previously defined with respect to the collar neighbourhoods, so for clarity let us refer to the topological manifold V' equipped with this new smooth structure as V. Then the map $\pi\colon V \to M$ is clearly smooth and carries the structure of a *smooth* rank 2 vector bundle. To obtain a complete metric on V with uniformly positive scalar curvature, we recall the following. The smooth structure on V' (defined via the collar neighbourhoods) is, up to diffeomorphism, the unique smooth structure on the topological manifold V', such that the topological embeddings

$$E \times [1,\infty) \to V', \;\; DE \to V'$$

are smooth embeddings [**?**, Thm. 1.4]. Note that the inclusion $E \times [1,\infty) \to V$ is locally given by the inclusion

$$U_\alpha \times (\mathbb{R}^2 - B) \to U_\alpha \times \mathbb{R}^2$$

with respect to the smooth trivialisations θ_α and ξ_α. Similarly, the inclusion $DE \to V$ is locally given by the inclusion

$$U_\alpha \times D^2 \to U_\alpha \times \mathbb{R}^2$$

with respect to ψ_α and ξ_α. Hence, we conclude from the uniqueness statement above that there exists a diffeomorphism $f\colon V \to V'$. Then the metric f^*h on V is complete and of uniformly positive scalar curvature. This completes the construction.

3 Further Questions

3.1 Problems with Extending the Construction to the Spin Case

The base spaces M of the circle bundles $E \to M$ constructed in the previous chapter are all totally non-spin manifolds. As we have seen in Chapter **??**, many constructions in positive scalar curvature geometry require the existence of a spin structure, so it may also be desirable to get examples of such circle bundles over spin manifolds. However, our methods do not straightforwardly extend to the spin case. In this section we discuss the difficulties that arise.

Suppose we are given an S^1-principal bundle $\pi\colon E \to M$ over a spin manifold M (in particular M is orientable). First of all, Theorem **??** no longer applies to conclude the existence of a PSC-metric on E, since the total space E is itself a spin manifold. This can be seen as follows. There exists a splitting

$$TE \cong VE \oplus HE,$$

of the tangent bundle TE of E into a *vertical* bundle VE and a *horizontal* bundle HE. The bundle VE is simply the kernel of the differential $d\pi\colon TE \to TM$, and we can then choose HE as any complementary subbundle (for example by choosing a fiber metric on TE). By construction, $d\pi\colon HE \to TM$ is an isomorphism on fibers, so that $HE \cong \pi^*TM$. In our case, VE is one-dimensional, so that $w_2(VE) = 0$ and therefore

$$w_2(TE) = w_1(HE) \smile w_1(VE) + w_2(HE) = \pi^* w_1(TM) \smile w_1(VE) + \pi^* w_2(TM) = 0,$$

using the fact that M is spin in the last equality. So instead, we are inclined to use the spin analogue of Theorem **??** to conclude the existence of a PSC-metric on E, namely:

Theorem ??. *Let W^{n+1} be a connected spin cobordism from M_0 to M_1, $n \geq 5$. Suppose M_1 is connected, the inclusion $M_1 \to W$ induces an isomorphism on π_1, and M_0 admits a* PSC*-metric. Then M_1 also admits a* PSC*-metric. In fact, any* PSC*-metric on M_0 can be extended to a* PSC*-metric on W, which is of product form near the boundary.*

Proceeding as before, we would like to take the disk bundle DE as a nullbordism of E and show:

(I) DE is spin,

(II) The inclusion $S^1 \to E$ is trivial on π_1.

However, these properties contradict each other. To see this, we make the following observations. Let $L \to M$ be the complex line bundle associated to $E \to M$ and consider the inclusion $\iota\colon DE \to L$. Then $TDE \cong \iota^*TL$ and thus

$$\iota^* w_2(TL) = w_2(TDE) = 0$$

by property (I). Since ι is a homotopy equivalence we obtain $w_2(TL) = 0$. Let $p\colon L \to M$ be the bundle projection. As before, we have a splitting

$$TL \cong VL \oplus HL$$

into a vertical and horizontal subbundle with $HL \cong p^*TM$. For vector bundles, it is a standard fact that $VL = \ker(dp) \cong p^*L$, hence

$$0 = w_2(TL) = p^*w_2(L) + p^*w_1(L) \smile p^*w_1(TM) + p^*w_2(TM) = p^*w_2(L).$$

Again, we conclude $w_2(L) = 0$ since p is a homotopy equivalence. Therefore, the first Chern class $c_1(L)$ lies in the kernel of the coefficient homomorphism

$$H^2(M;\mathbb{Z}) \to H^2(M;\mathbb{Z}_2)$$

by Proposition 1.17. In other words, $c_1(L)$ can be written as $c_1(L) = 2\alpha$ for some $\alpha \in H^2(M;\mathbb{Z})$. To derive a contradiction to property (II) from this, we claim that the following diagram commutes up to sign

$$\begin{tikzcd}
\pi_2(M) \arrow[r, "\partial"] \arrow[d] & \pi_1(S^1) \arrow[d, "\cong"] \\
H_2(M;\mathbb{Z}) \arrow[r, "{\langle c_1(L), - \rangle [S^1]}"] & H_1(S^1;\mathbb{Z})
\end{tikzcd} \tag{3.1}$$

where ∂ is the boundary map from the long exact sequence of homotopy groups for the fibration $E \to M$ and the vertical maps are given by the Hurewicz homomorphism. Moreover, $[S^1]$ denotes the fundamental class of S^1. Observe that $\pi_1(S^1) \to H_1(S^1;\mathbb{Z})$ is an isomorphism, since $\pi_1(S^1)$ is abelian. Granting this claim for now, it follows from $c_1(L) = 2\alpha$ that the image of $\partial\colon \pi_2(M) \to \pi_1(S^1)$ contains only even multiples of the generator of $\pi_1(S^1)$. Thus ∂ is not surjective, and consequently $\pi_1(S^1) \to \pi_1(E)$ is non-zero by exactness. Therefore, a bundle satisfying both (I) and (II) does not exist. To show commutativity of the above square, let $\tau_1 \to \mathbb{C}P^\infty$ be the universal complex line bundle (compare Example 1.14 (vi)) and let $f\colon M \to \mathbb{C}P^\infty$ be a classifying map for $L \to M$. Essentially, commutativity of the square is then due to the fact that the corresponding square for the *universal* principal bundle commutes and all maps are compatible with the classifying map f. To make this precise, we extend the diagram above as follows:

$$\begin{tikzcd}
\pi_2(\mathbb{C}P^\infty) \arrow[rrr, bend left, "\partial", "\cong"'] \arrow[ddd, "\cong"'] \arrow[rrr, phantom, shift right=4ex, "(2)"] & & & \pi_1(S^1) \arrow[ddd, "\cong"] \\
& \pi_2(M) \arrow[r, "\partial"] \arrow[d] \arrow[ul, "f_*"'] \arrow[dl, phantom, "(3)"] & \pi_1(S^1) \arrow[d, "\cong"] \arrow[ur, "\mathrm{id}"] \arrow[dr, phantom, "(1)"] & \\
& H_2(M;\mathbb{Z}) \arrow[r, "{\langle c_1(L), - \rangle [S^1]}"] \arrow[dl, "f_*"] & H_1(S^1;\mathbb{Z}) \arrow[dr, "\mathrm{id}"] & \\
H_2(\mathbb{C}P^\infty;\mathbb{Z}) \arrow[rrr, bend right, "{\langle c_1(\tau_1), - \rangle [S^1]}"] \arrow[rrr, phantom, shift left=4ex, "(4)"] & & & H_1(S^1;\mathbb{Z})
\end{tikzcd}$$

Again, the top horizontal map is the boundary map from the long exact sequence for the fibration $ES^1 \to \mathbb{C}P^\infty$ and the outer vertical maps are given by the Hurewicz homomorphism. Since $\mathbb{C}P^\infty$ is simply connected, $\pi_2(\mathbb{C}P^\infty) \to H_2(\mathbb{C}P^\infty;\mathbb{Z})$ is an isomorphism. Moreover, the boundary map $\partial\colon \pi_2(\mathbb{C}P^\infty) \to \pi_1(S^1)$ is an isomorphism by Example 1.14 (i). Clearly, (1) commutes. Furthermore, (2) commutes by naturality of the long exact sequence of homotopy groups and (3) commutes by naturality of the Hurewicz homomorphism. For commutativity of (4), we compute for $a \in H_2(M;\mathbb{Z})$:

$$\langle c_1(\tau_1), f_*a \rangle = \langle f^*c_1(\tau_1), a \rangle = \langle c_1(L), a \rangle,$$

using the fact that f is a classifying map for L in the last equation. Finally, we have to show commutativity of the outer square. From the universal coefficient theorem, we obtain that

$$H^2(\mathbb{C}P^\infty;\mathbb{Z}) \to \mathrm{Hom}(H_2(\mathbb{C}P^\infty;\mathbb{Z}),\mathbb{Z}), \quad \beta \mapsto \langle \beta, - \rangle$$

is an isomorphism. By definition, $c_1(\tau_1) \in H^2(\mathbb{C}P^\infty;\mathbb{Z})$ is a generator. Thus $\langle c_1(\tau_1), - \rangle$ is a generator of $\mathrm{Hom}(H_2(\mathbb{C}P^\infty;\mathbb{Z}),\mathbb{Z})$ and consequently evaluates to ± 1 on a generator of $H_2(\mathbb{C}P^\infty;\mathbb{Z})$. From this we see that

$$\langle c_1(\tau_1), - \rangle [S^1] \colon H_2(\mathbb{C}P^\infty;\mathbb{Z}) \to H_2(S^1;\mathbb{Z})$$

is an isomorphism. So all maps in the outer square are isomorphisms and, moreover, all groups are isomorphic to $\mathbb{Z}$. Since the only isomorphisms of $\mathbb{Z}$ are $\pm\mathrm{id}$, the outer square commutes up to sign. It follows that the inner square commutes as well.

Thus, we have established that our previous methods for constructing a PSC-metric on $E \to M$ do not apply when M is a spin manifold. Nevertheless, examples in the spin case do exist. These were obtained by Kumar and Sen [**?**] using more sophisticated methods, as we have briefly sketched in the Introduction.

3.2 A Question about the Homotopy Groups of the Space of PSC-Metrics

In Chapter 2 we have discussed sufficient conditions for the existence of a PSC-metric on a manifold. Given a manifold M which admits a PSC-metric, we can further ask how many such metrics exist. More precisely, the question should be whether any two PSC-metrics on M can be deformed into each other via a continuous path of PSC-metrics. Phrased differently, if we denote by $\mathcal{R}^+(M)$ the space of all PSC-metrics on M, equipped with the C^∞-topology, this statement asks whether $\pi_0(\mathcal{R}^+(M))$ is trivial. For instance, this is always true for $M = S^2$. In fact, $\mathcal{R}^+(S^2)$ is contractible, and so in particular connected [**?**, Thm. 3.4]. However, in higher dimensions this breaks down even for such simple spaces as spheres, as demonstrated by the following theorem due to Carr [**?**, Thm. 4].

Theorem 3.1. *For $n \geq 2$ the space $\mathcal{R}^+(S^{4n-1})$ has infinitely many path components.*

This already illustrates the complexity that the space of PSC-metrics may exhibit. Going one step further, we can also investigate the higher homotopy groups of this space, essentially asking which *families* of PSC-metrics can be deformed into each other. One result in this direction is the following theorem due to Hanke, Schick and Steimle [**?**, Thm. 1.1].

Theorem 3.2. *Let $k \in \mathbb{N}_0$. Then there exists a number $N(k)$ such that for every $n \geq N(k)$ and every spin manifold M of dimension $4n - k - 1$ admitting a* PSC*-metric g_0, the homotopy group*

$$\pi_k(\mathcal{R}^+(M), g_0)$$

contains elements of infinite order if $k \geq 1$ and infinitely many distinct elements if $k = 0$.

Note that for $k = 0$ and $M = S^{4n-1}$, the statement above generalizes Theorem 3.1 quite substantially. Tying this discussion back to our results of Chapter 4, let us analyze the PSC-metric constructed on E more closely. A natural question is whether this metric is invariant under the S^1-action on E. We can answer this question in the negative due to a theorem of Bérard-Bergery, which states that a closed manifold with a free S^1-action admits an S^1-invariant PSC-metric if and only if its orbit space admits a PSC-metric [**?**, Theorem C], [**?**, Theorem 2.2]. In our case, $E/S^1 \cong M$ and M is enlargeable. Therefore, it follows from Theorem **??** that M does not admit a PSC-metric, and hence the PSC-metric on E cannot be S^1-invariant. In view of the above discussion about the homotopy groups of spaces of PSC-metrics, we are led to the following question.

Question. Let $g \in \mathcal{R}^+(E)$ be the PSC-metric constructed in Chapter 4. Does the map

$$S^1 \to \mathcal{R}^+(E), \;\; \zeta \mapsto \zeta^* g$$

represent a non-trivial element in $\pi_1(\mathcal{R}^+(E), g)$?

Here we denote by $\zeta^* g$ the PSC-metric on E arising as the pullback of g under the diffeomorphism induced by $\zeta \in S^1$. Given the preceding observations, the most that can be said is that this loop is non-constant. However, it may of course still be contractible. At the time of writing, we do not know the answer to the above question and leave it as an open problem for future investigation.